PENGZHEN LI

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EDUCATION

University of Tennessee, Knoxville expected May 2027
Ph.D. in Economics

Shandong University, Jinan 2018
M.A. in Finance — thesis published in *Technological Forecasting and Social Change* (2019); 1,000+ citations

Tongji University, Shanghai 2015
B.A. in Economics

RESEARCH INTERESTS

Environmental and Energy Economics, Agricultural and Resource Economics, Applied Econometrics, and International Trade.

PEER-REVIEWED PUBLICATIONS

Four SSCI/SCI-indexed publications; 1,132 total citations, h-index 4 (Google Scholar, August 2026).

"Do green technology innovations contribute to carbon dioxide emission reduction? Empirical evidence from patent data." K. Du, **P. Li**, and Z. Yan. *Technological Forecasting and Social Change*, 146 (2019), 297–303. **1,000+ citations (Google Scholar).**

"Spatial optimization of the sustainable aviation fuel supply chains from forest residues via fast pyrolysis/hydrotreatment considering feedstock ash content variability." **P. Li**, T. E. Yu, N. Labbé, N. Abdoulmoumine, M. Garcia-Perez, K. P. Hoyt, and B. C. English. *Biomass and Bioenergy*, 208 (2026), 108793.

"Assessing the impact of preprocessing and conversion technologies on the sustainable aviation fuel supply from forest residues in the Southeast USA." **P. Li**, T. E. Yu, C. Trejo-Pech, J. A. Larson, B. C. English, and D. N. Lanning. *Transportation Research Record*, 2678(9) (2024), 639–654.

"Beef price spread relationship with processing capacity utilization." C. Martinez, **P. Li**, C. N. Boyer, T. E. Yu, and J. G. Maples. *Journal of the Agricultural and Applied Economics Association*, 2(1) (2023), 133–145.

WORKING PAPERS

Selling Down-Market: Market Access and the Quality of Chinese Exports [JMP] [paper]
Abstract: Trade liberalization can change export quality through cheaper inputs or through access to foreign buyers. I study the latter, exploiting the 2001 US grant of permanent normal trade relations to China, which eliminated the threat of tariff increases that varied across products. Comparing the same firm's US and non-US sales of a product before and after 2002, I find that quality fell about 4.5 percent more in highly exposed products than in barely exposed ones. The fall appears on impact, does not detectably reverse, and has no counterpart in the EU. Prices fell, but quantities rose less than the price decline alone predicts. I develop a model in which access lets firms sell down-market, to buyers who value quality less.

Trade Restrictiveness Indices with Discriminatory Tariffs with James Lake
Abstract: A Trade Restrictiveness Index (TRI) is the uniform tariff that is welfare-equivalent to a country's actual tariff structure. Its standard implementation assigns a single tariff to each product, yet preferential trade agreements make tariffs on the same product differ across exporters, distorting where a country buys as well as what it buys. In an Armington model, we derive a TRI that separates these two distortions: for a small country, the squared index is, to second order, the square of the conventional index evaluated at mean product tariffs plus a term in the variance of tariffs across each product's exporters. We also derive the index for large importers, whose tariffs affect world prices. Across 123 importers over 2000–2020, the dispersion term accounts for 24 percent

of the median importer's squared TRI in 2018, up from zero in the early 2000s. Preferences therefore reduce the TRI far less than they reduce average tariffs: in 2018 they cut the median importer's simple average tariff by 16 percent but its TRI by only 5 percent.

The Impact of Mayoral Endorsements on Presidential Election Outcomes: Insights from Brazil with Matt Van Essen and Luiz Lima. *2024 J. Fred and Wilma Holly Award (best second-year paper).*
Abstract: Brazilian mayors' endorsements are valuable to presidential candidates, and a mayor's return from endorsing depends on whom other mayors endorse, because the budgetary prize controlled by the winner is shared among the mayors who backed him. We model endorsements as a static discrete-choice game of incomplete information in which mayors act on public information and private preference shocks. With extreme-value shocks and the payoff from not endorsing normalized to zero, equilibrium endorsement probabilities map uniquely back to expected payoffs, and an exclusion restriction on payoffs identifies the structural parameters. We estimate the game by maximum likelihood with equilibrium constraints, which avoids computing every equilibrium at each trial parameter value. In Monte Carlo experiments the estimator recovers the structural parameters to within 1 percent and the equilibrium probabilities to within 0.1 percent. An application to a new textual dataset on Brazilian mayors' endorsements is planned.

WORK IN PROGRESS

Trade Restrictiveness Indices 2.0 with Massimiliano Cali, Aufa Doraset, James Lake and Laura Puzzello

RESEARCH EXPERIENCE

Graduate Research Assistant, University of Tennessee 2021 – 2023
USDA NIFA (two projects) and FAA Office of Environment and Energy (ASCENT) funded research on forest-residue biofuel and sustainable aviation fuel (SAF) supply chains in the Southeast U.S.

- Designed cost-minimizing and profit-maximizing supply-chain networks with mixed-integer programming (GAMS); results published in *Biomass and Bioenergy* and *Transportation Research Record*.
- Facility-siting and investment analysis combining econometric models with the IMPLAN regional impact platform; efficiency analysis of facilities and airports with Data Envelopment Analysis.
- Sensitivity and scenario analysis of feedstock quality and market conditions with extended mathematical programming (EMP), supervised machine learning, and @RISK.
- Managed high-resolution geospatial data in ArcGIS, R, and Python.

Other research experience

- *Panel and causal methods:* synthetic-control estimate of the long-run output effect of the 2008 Wenchuan earthquake; GMM estimates of MFP/CFAP payments on U.S. agricultural lending; Hansen panel-threshold models of green innovation and CO_2; mediation analysis of digital finance and emissions for Chinese listed firms.
- *Time series:* threshold VAR of regime switching in commodity-price linkages; VECM and multivariate GARCH of wholesale commodity-price volatility.
- *Machine learning:* drivers of CO_2 emissions from patent data; equity-premium prediction from *NYT* and *WSJ* text.

PRESENTATIONS

Western Regional Science Association, Annual Meeting, Hawaii 2023
Western Agricultural Economics Association, Annual Meeting, Santa Fe, NM 2022
Southern Agricultural Economics Association, Annual Meeting, New Orleans, LA 2022
Midwest Economic Theory and International Trade Meetings, Virginia Tech Spring 2025
University of Tennessee, Economics Brown Bag Seminar Series Spring 2024, Fall 2024
University of Tennessee, Applied Micro Group Spring 2024, Fall 2024

TEACHING EXPERIENCE

University of Tennessee

Instructor

ECON 351: Monetary Economics	Spring 2025, Spring 2026
ECON 322: The Global Economy: Trade and Development	Summer 2025
ECON 313: Intermediate Macroeconomics	Summer 2026

Teaching Assistant

Ph.D. courses:

ECON 511: Microeconomic Theory	Fall 2023, Fall 2024, Fall 2025
ECON 512: Advanced Microeconomics	Spring 2024
ECON 581: Mathematical Methods in Economics	Summer 2024

Master's courses:

AREC 525: Agribusiness Operations Research Methods	Spring 2022, Spring 2023

HONORS AND AWARDS

Charles B. Garrison Award for Excellence in Teaching (best teaching), University of Tennessee	2026
Summer Graduate Assistantship, University of Tennessee	2025
J. Fred and Wilma Holly Award (best second-year paper), University of Tennessee	2024
J. Fred and Wilma Holly Fellowship, University of Tennessee	2023 – present
Graduate Assistantship, University of Tennessee	2021 – present
Academic Scholarship for Master's Students, Shandong University	2017
Freshman Scholarship, Shandong University	2016
Outstanding Rainbow Volunteer, Tongji University	2013 – 2015
National Endeavor Scholarship, Tongji University	2014
Tongji Scholarship of Excellence, Third Prize, Tongji University	2014

SKILLS

Programming Languages: Stata, R, Python, SQL, GAMS, MATLAB, LaTeX, ArcGIS.
Methods: panel and causal econometrics (FE, IV, DiD, event study, SCM, GMM, panel threshold), time series (TVAR, VECM, mGARCH), mixed-integer and stochastic programming, DEA, cooperative games, discrete-choice game estimation, machine learning.
Credentials: CFA Level II.
Languages: Chinese (native), English (fluent).

REFERENCES

T. Edward Yu
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